

FANNO-Dev: 167 Figures, 153K Samples — Complete Analysis Summary

Key Metrics

Samples	153,351
Figures	167
Tables	27
Reports	60+
Domains	2,297
Vendi	182.75

Pipeline Rankings

- 1. Self-Inversion**
Most diverse, efficient, domains
- 2. Document QA**
Best LOO value, high Vendi
- 3. Complex QA**
Largest source, broad coverage
- 4. Reasoning QA**
Best teamwork, Q-A alignment
- 5. Code QA**
Most frontier samples
- 6. Math QA**
Distinct semantic niche
- 7. Creative**
Structural uniqueness

Top 8 Findings

- Best pair: Document QA + Self-Inversion (Vendi=135.1, synergy=+31.9)
- Greedy order: Self-Inv -> Doc QA -> Complex -> Reasoning -> Code -> Math -> Creative
- Self-Inversion: 2,288 domains (99.1% exclusive), 87.6% hull coverage, all 20 K-Means clusters
- Scaling law: $V(N) = 55.4(1-\exp(-N/745))+109.0$, $R^2=0.996$, ceiling ~164.5
- All length distributions follow Log-Normal; Math best fit (KS=0.037)
- Cost: \$2,245 total (\$15.58/1K); Self-Inv most cost-efficient (Vendi/\$=29.4)
- Quality: all pipelines >95% score; 0 benchmark contamination
- Cross-source distance: 0.961 average (nearly orthogonal pipelines)

Session 6 Analyses (fig138-167)

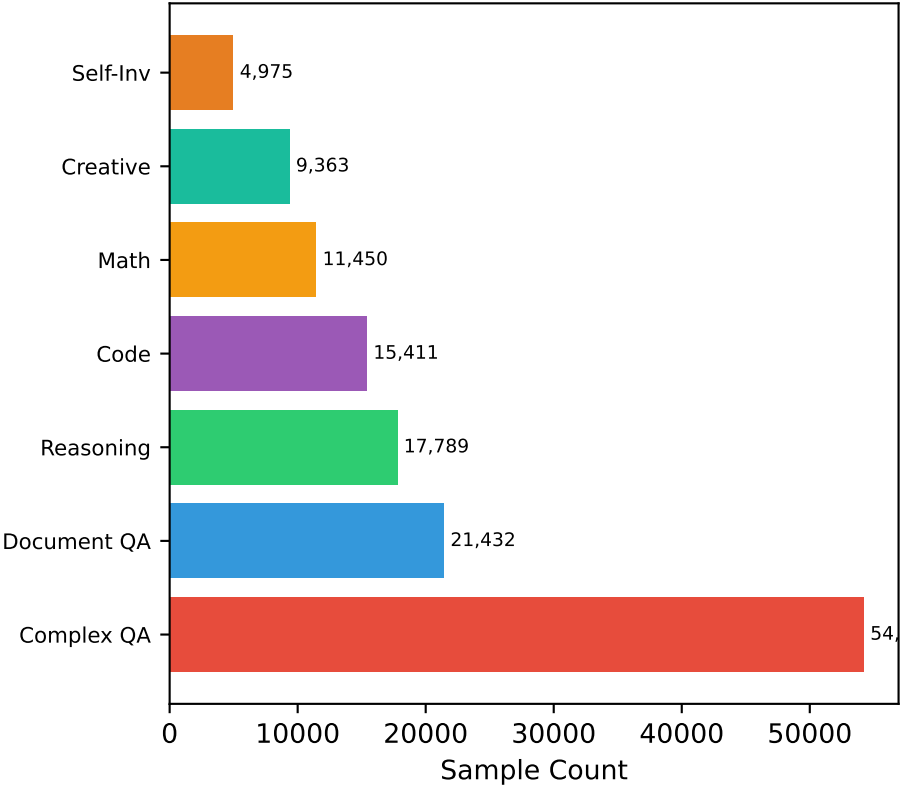
- fig139: Length Distribution Modeling**
-> All Log-Normal
- fig140: Boundary Analysis**
-> Self-Inv 87.6% hull
- fig141: Topic Coherence (NMF)**
-> 15 specialized topics
- fig142: Transfer Analysis**
-> Self-Inv irreplaceable
- fig143: Eigenspectrum**
-> Self-Inv eff_dim=61.6
- fig148: Pairwise Diversity**
-> DocQA+SI best (135.1)
- fig150: Paper Highlights**
-> 9-panel overview

- fig152: Greedy Incremental**
-> First 2 get 64%
- fig156: Clustering Stability**
-> K=30 best silhouette
- fig158: Semantic Bridges**
-> Math most isolated
- fig160: Paper Figure 1**
-> Publication candidate
- fig165: Self-Inv Deep Dive**
-> 2,064 exclusive domains
- fig166: Quality Scorecard**
-> All >95% quality

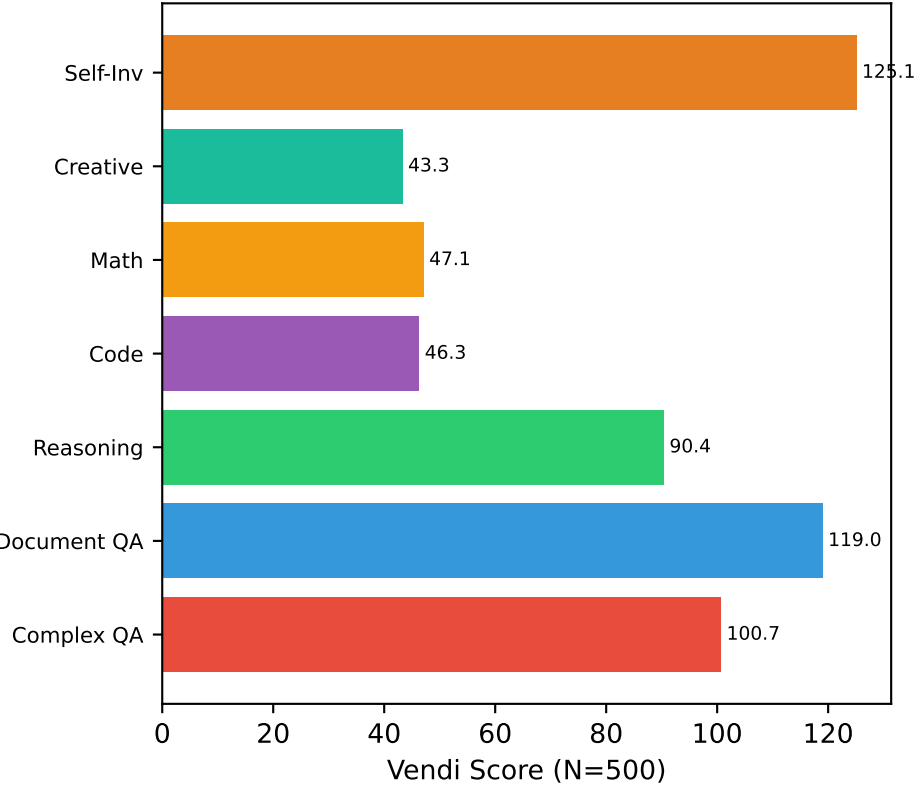
Deliverables Inventory

PNG+PDF Figures:	167
LaTeX Tables:	27
JSON Reports:	60
Paper Draft:	1
Git Commits:	150

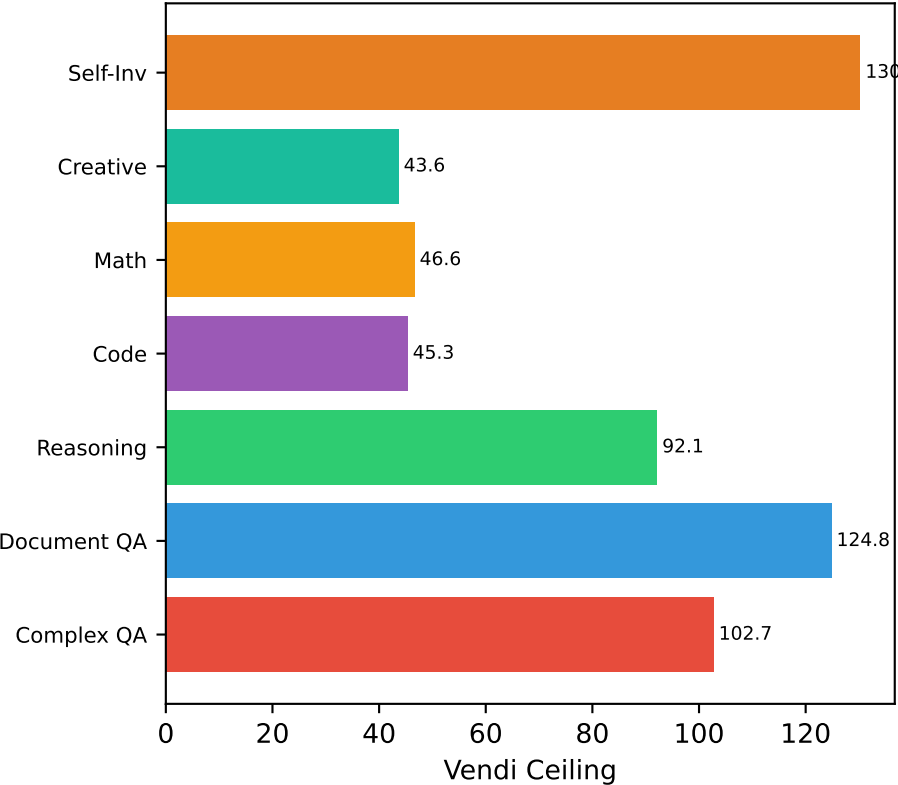
Samples per Pipeline



Per-Pipeline Diversity



Saturation Ceiling



Quality Score

